

DEVOPS

Development

From writing code to reliable delivery

B.Tech CSE Presentation



What is DevOps?

02

DEV

Development

Code • Build • Test

OPS

Operations

Deploy • Run • Monitor

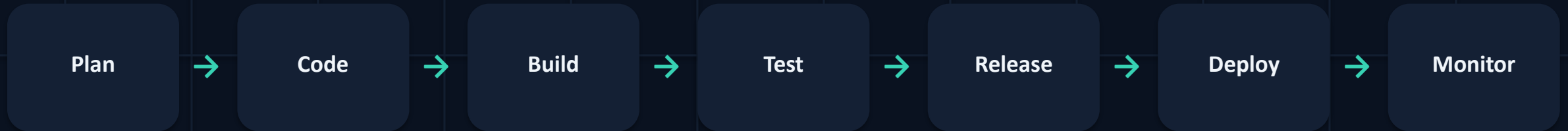
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DevOps

Collaboration + Automation

Development in the DevOps Lifecycle

03



Development connects coding with automation, testing and delivery.

Role of a Developer

04

01

Write Code

02

Use Git

03

Automate Tests

04

Review & Fix

05

Collaborate

Git & GitHub – Version Control

05

LOCAL REPOSITORY

Working Tree

Commit

Branch



GITHUB

Push

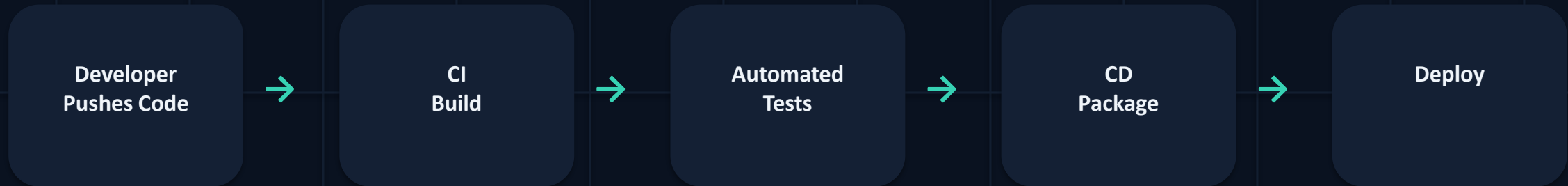
Pull Request

Code Review

Merge

CI/CD – Continuous Integration & Delivery

06



Automation reduces manual errors and provides fast feedback.

Testing & Code Quality

07

Unit Test

Checks individual functions

✓
✓
✓

Integration Test

Checks components together

✓
✓
✓

Quality Scan

Finds bugs & vulnerabilities

✓
✓
✓

Docker & Containerization

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APPLICATION

App Code

Libraries

Dependencies

Configuration

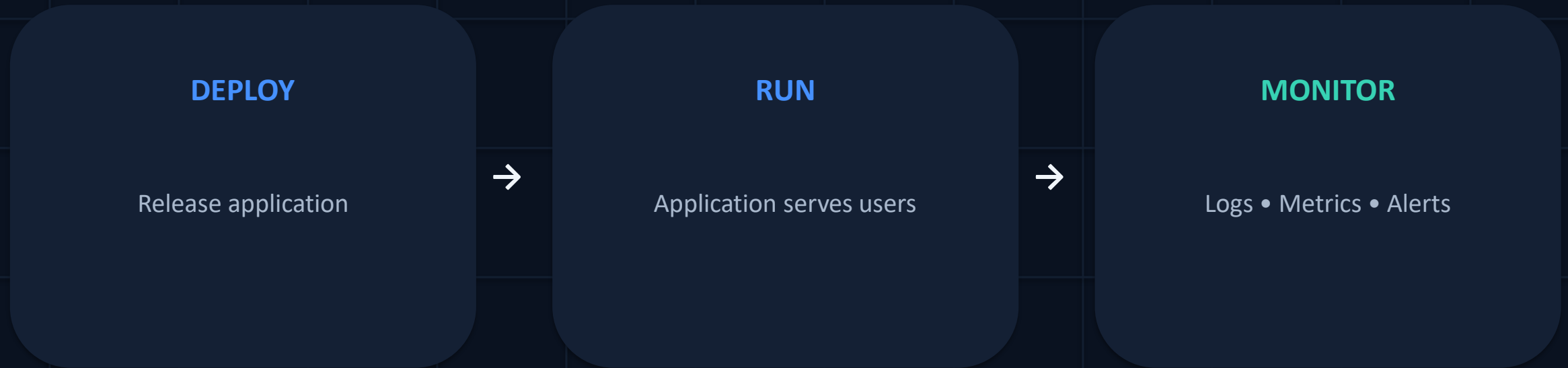


CONTAINER

**Same application
+ dependencies
= consistent environment**

Deployment & Monitoring

09



Benefits of DevOps Development

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✓ **Faster Releases**

✓ **Better Collaboration**

✓ **Early Bug Detection**

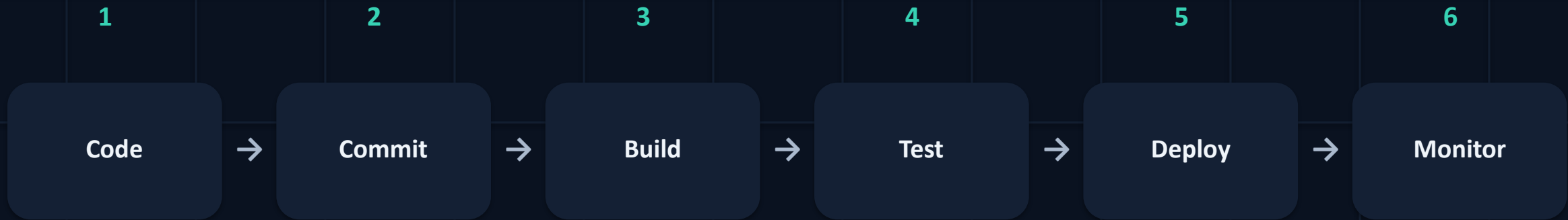
✓ **Reliable
Deployments**

✓ **Less Manual Work**

✓ **Continuous Feedback**

A Simple DevOps Workflow

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Feedback from monitoring goes back into development.

Development is the engine of DevOps.

**Code + Git + CI/CD + Testing + Containers
help teams deliver software faster, safer and more reliably.**

Thank You